Temple Syndrome (TS14): A Comprehensive Disease Characteristics Report

Disease: Temple syndrome (TS14) **MONDO ID:** MONDO:0100507 · **OMIM:** #616222 · **Orphanet:** ORPHA:254528 **Causal locus:** Chromosome 14q32.2 imprinted *DLK1-DIO3* domain **Category:** Mendelian / imprinting disorder

Summary

Temple syndrome is a rare genomic imprinting disorder of the chromosome 14q32.2 *DLK1-DIO3 domain. Its defining molecular lesion is the loss of expression of the paternally expressed, protein-coding genes — chiefly DLK1 (Delta-Like Non-Canonical Notch Ligand 1), together with RTL1 and DIO3* — combined with biallelic over-expression of the maternally expressed non-coding RNAs of the same cluster (*MEG3*/GTL2, *MEG8*, the *RTL1* antisense transcript, and an embedded miRNA/snoRNA array). This dosage imbalance is produced by three convergent mechanisms, all of which manifest as hypomethylation of the *MEG3/DLK1* intergenic differentially methylated region (IG-DMR) and the *MEG3*-DMR: (1) **maternal uniparental disomy of chromosome 14 [upd(14)mat]**, (2) **isolated paternal epimutation (loss of methylation)** of the IG-DMR/MEG3-DMR, and (3) **paternal microdeletion of 14q32**. In the two largest cohorts the proportions are roughly 29–61% epimutation, 29–88% upd(14)mat, and 5–11% deletion, varying with ascertainment [PMID: 29659920;

39693239

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Clinically, TS14 presents as a multi-system growth and metabolic disorder. Cardinal features are pre- and postnatal growth restriction (most children are small for gestational age), neonatal hypotonia and feeding difficulties, motor and language delay, small hands and feet, later truncal obesity, and **central precocious puberty (CPP)** — the last driven specifically by loss of *DLK1*. Pooled data from 51 published cases gave hypotonia 93%, motor delay 83%, early puberty 86% (mean menarche 10 years 2 months), small hands 87%, small feet 96%, and

obesity 49% [PMID: 24891339]. A genetically confirmed 60-patient Japanese cohort reported SGA 88.3%, postnatal short stature 87.0%, and CPP 86.0% [PMID: 39693239]. Cognition is generally normal to mildly reduced (median IQ ~85–91), and adult functional outcome is favourable (98.2% attended college or held jobs), but there is meaningful lifelong cardiometabolic risk — obesity ~20%, hypercholesterolemia ~27%, and early type 2 diabetes ~13% [PMID: 39693239;].

P 36362517

TS14 is clinically important chiefly because it is under-diagnosed and treatable. It overlaps substantially with Silver-Russell syndrome (SRS) and Prader-Willi syndrome (PWS) and is identified in roughly 2% of patients referred for PWS or SRS testing [PMID: 29159982]. Diagnosis rests on tiered methylation and copy-number/UPD testing of 14q32. Management is supportive and multidisciplinary: recombinant growth hormone (rhGH) improves short- and long-term height and body composition, GnRH-analogue therapy suppresses precocious puberty, and lifelong metabolic surveillance addresses the obesity/dyslipidemia/diabetes risk. No disease-modifying or gene-directed therapy exists. Recurrence risk is low for the common sporadic mechanisms, but paternally inherited 14q32 deletions are Mendelian with a parent-of-origin twist: the *same* deletion transmitted through a mother produces the mirror-image disorder, **Kagami-Ogata syndrome (KOS)**.

Section 1 — Disease Information

Overview. Temple syndrome (TS14) is a rare, clinically recognizable imprinting disorder caused by absence of the paternally expressed genes and/or over-expression of the maternally expressed non-coding RNAs within a single imprinted cluster on human chromosome 14q32 [PMID: 32592473]. It was delineated as a distinct entity and given its eponym by Ioannides et al. in 2014 after analysis of 51 published cases of what had previously been called the "maternal UPD14 phenotype," honouring clinician-geneticist I. Karen Temple [PMID: 24891339].

Key identifiers.

Resource	Identifier
OMIM	#616222
MONDO	MONDO:0100507
Orphanet	ORPHA:254528
HGNC genes	*DLK1* (HGNC:2907), *MEG3* (HGNC:14575), *RTL1* (HGNC:35814), *MEG8*, *DIO3*

Synonyms / alternative names: maternal uniparental disomy 14 syndrome; upd(14)mat phenotype/syndrome; TS14; 14q32.2 maternal imprinting defect [PMID: 34460908; PMID: 39693239].

Information source. This report is derived from aggregated disease-level resources (OMIM, Orphanet, MONDO) and from published clinical cohorts and case series (human clinical evidence), supplemented by model-organism and in-vitro mechanistic studies. It is not EHR/individual-patient derived.

Section 2 — Etiology

Primary cause — genetic/epigenetic. TS14 is a constitutional (epi)genetic disorder with **no infectious, toxic, or classical lifestyle cause.** The primary lesion is a parent-of-origin dosage imbalance at 14q32.2 produced by three mechanisms [PMID: 38715103; PMID: 32592473; PMID: 29659920]:

1. **Maternal uniparental disomy of chromosome 14 [upd(14)mat]** — typically a maternal meiosis-I nondisjunction error followed by postzygotic trisomy-14 rescue (occasionally monosomy rescue or a mitotic error), which explains the occasional co-occurrence of TS14 with mosaic trisomy 14 [PMID: 30181735; PMID: 33746039].

2. **Isolated primary epimutation** — sporadic loss of methylation at the paternal IG-DMR/MEG3-DMR [PMID: 26395259; 25351781].
3. **Paternal 14q32 microdeletion** — heritable and Mendelian [PMID: 38715103].

Genetic risk factors. The causal "variants" are the (epi)genetic lesions above. There are no established common susceptibility loci or modifier alleles with proven effect on TS14 risk. Multilocus imprinting disturbance (MLID) can co-affect other imprinted loci (e.g., 11p15), sometimes attributable to variants in methylation-machinery genes, and represents an etiologic modifier rather than an independent cause [PMID: 35266280].

Environmental risk factors. (1) **Advanced maternal age** is associated with the meiotic nondisjunction that underlies upd(14)mat via trisomy rescue [PMID: 30181735]. (2) **Assisted reproductive technology (ART)** is a recognized (though not TS14-specific) risk for imprinting epimutations: "ART procedures, especially ovarian hyperstimulation, in vitro fertilization (IVF), and embryo culture, coincide with critical periods of epigenetic reprogramming and may contribute to epimutations in imprinting control regions" [PMID: 41153459].

Protective factors. No protective alleles or environmental protective factors are established for TS14.

Gene–environment interactions. The *DLK1-MEG3* methylation state is environmentally sensitive: preeclampsia exposure is associated with hypermethylation of the IG-DMR and altered *DLK1/MEG3* expression in umbilical vessels, illustrating that environmental stress can perturb this locus [PMID: 30234781]. This is mechanistically illustrative rather than a demonstrated cause of TS14.

Section 3 — Phenotypes

TS14 phenotypes are predominantly physical manifestations, clinical signs, and some behavioural/neurodevelopmental features, with a few laboratory abnormalities (dyslipidemia, hyperglycemia) emerging later in life.

Phenotype	HPO term (suggested)	Type	Onset	Frequency	Source
Intrauterine / prenatal growth restriction	HP:0001511	Physical	Prenatal	67–88% SGA; 100% IUGR in some cohorts	P 39693239; P 36362517
Postnatal short stature	HP:0004322	Physical	Childhood (~2 y)	87%	P 39693239
Muscular hypotonia	HP:0001252	Clinical sign	Neonatal	71–100% (pooled 93%)	P 24891339; P 36362517
Feeding difficulties	HP:0011968	Clinical sign	Neonatal	53–86%	P 42026594; P 36362517
Motor delay	HP:0001270	Behavioural/ neuro	Infancy	83%	P 24891339
Delayed speech / language	HP:0000750	Behavioural/ neuro	Childhood	Common	P 42026594
Small hands	HP:0200055	Physical	Congenital	87%	P 24891339
Small feet	HP:0001773	Physical	Congenital	96%	P 24891339
Truncal obesity / weight gain	HP:0001956	Physical	Later childhood	~20–49%	P 24891339; P 39693239
Central precocious puberty	HP:0000826	Clinical sign	Childhood	86–89%	P 39693239; P 36362517
	HP:0001249	Behavioural/ neuro	Childhood		

Phenotype	HPO term (suggested)	Type	Onset	Frequency	Source
Intellectual/ developmental disability (mild)				~22% (IQ median ~85–91)	P 39693239; P 36362517
Hypercholesterolemia	HP:0003124	Lab abnormality	≥6 y	26.5%	P 39693239
Diabetes mellitus (type 2)	HP:0000819	Lab abnormality	≥9 y	12.8%	P 39693239

Severity/progression. Core features (growth restriction, small hands/feet, hypotonia) are congenital and largely stable/non-progressive. The metabolic phenotype (obesity, dyslipidemia, early diabetes) is progressive/age-dependent. Precocious puberty is episodic in presentation but reversible with treatment. Expressivity is variable.

Quality-of-life impact. TS14 affects growth, feeding, motor/language development, and psychosocial function; psycho-behavioral problems were reported in ~53% of one cohort [PMID: 36362517]. Despite early special-education enrolment (~43–54%), the great majority achieve independent adult function [PMID: 39693239]. No TS14-specific EQ-5D/SF-36 data are available.

Section 4 — Genetic / Molecular Information

Causal genes and domain architecture. The disorder maps to the imprinted *DLK1-DIO3* domain at 14q32.2. "The domain contains three imprinted protein-coding genes, Dlk1, Rtl1 and Dio3, expressed from the paternally inherited chromosome and several imprinted large and small noncoding RNA genes expressed from the maternally inherited homolog" [PMID: 18471925]. Loss of the paternal program (DLK1, RTL1, DIO3) plus gain of the maternal program (MEG3/GTL2, MEG8, RTL1as, miRNAs/snoRNAs) produces TS14 [PMID: 38715103].

- **DLK1** (HGNC:2907; OMIM *176290) — the principal driver of the precocious-puberty and adiposity phenotype. "Loss-of-function mutations in DLK1 … have been linked to central precocious puberty (CPP) and increased body fat in girls" [PMID: 42677749].

- **MEG3/GTL2** (HGNC:14575), **MEG8**, **RTL1/RTL1as** (HGNC:35814), **DIO3** — maternally or paternally expressed cluster members contributing to the imbalance.

Pathogenic variants / lesion classes.

Mechanism	Lesion class	Approx. frequency (Geoffron, n=28)	Consequence
IG-DMR/MEG3-DMR epimutation	Primary epigenetic LOM	60.7%	Silences paternal DLK1/RTL1
upd(14)mat	Whole-chromosome UPD (structural)	28.6%	Loss of paternal alleles
Paternal 14q32 microdeletion	Structural deletion	10.7%	Loss of paternal copy

"Seventeen (60.7%) patients showed loss of methylation of the MEG3/DLK1 intergenic differentially methylated region by epimutation. Eight (28.6%) patients had maternal uniparental disomy of chromosome 14 and three (10.7%) had a paternal deletion in 14q32.2" [PMID: 29659920]. Proportions differ by cohort and referral pattern; in the 60-patient Japanese cohort upd(14)mat predominated (31/60) [PMID: 39693239].

Variant classification / origin. The lesions are germline/constitutional (not somatic). Deletions are classed pathogenic; epimutations and UPD are pathogenic epigenetic/structural events rather than ACMG sequence variants. Allele-frequency databases (gnomAD etc.) are not applicable to UPD/epimutation. A rare novel structural mechanism — **interchromosomal template-switching** forming DUP-TRP/INV-DUP complex rearrangements and generating segmental UPD(14)mat — has been described: "these CGRs may lead to genetic abnormalities in carriers due to dosage-sensitive genes mapping within the copy-number variant regions" [PMID: 31014393].

Functional consequence. Net effect is **loss of function of paternally expressed genes** (DLK1/RTL1/DIO3) with **gain of expression of maternally expressed non-coding RNAs**.

Modifier genes. No robust modifier genes are established. MLID can co-involve machinery genes affecting multiple loci [PMID: 35266280]. In SRS, pathogenic *PLAG1* variants secondarily deregulate the MEG8:Int2-DMR of the 14q32 cluster, connecting this domain to a wider growth-disorder network [PMID: 41276848].

Epigenetic information. The germline **IG-DMR** (methylated on the paternal allele) hierarchically controls imprinting across the domain, with the somatic **MEG3-DMR** downstream; paternal loss of IG-DMR methylation is the unifying biomarker of TS14 [PMID: 18471925;

25351781

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Chromosomal abnormalities. upd(14)mat (± mosaic trisomy 14 or small supernumerary marker chromosome 14) and paternal 14q32 microdeletions [PMID: 33746039;

30181735

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Section 5 — Environmental Information

TS14 has **no established toxic, occupational, pollution, dietary, or infectious cause.** Environmental relevance is confined to (a) ART-associated epimutation risk during epigenetic reprogramming windows [PMID: 41153459;

39889767

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and (b) demonstration that the *DLK1-MEG3* methylation state is modifiable by gestational stress such as preeclampsia [PMID: 30234781]. No infectious agents are implicated.

Section 6 — Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. A parent-of-origin lesion at 14q32.2 occurs — **upd(14)mat**, **paternal IG-DMR/MEG3-DMR epimutation**, or **paternal 14q32 microdeletion** — *leads to* loss of the paternal allele's contribution across the domain [PMID: 38715103].
2. Loss of paternal IG-DMR methylation *results in* silencing of the paternally expressed protein-coding genes **DLK1, RTL1, DIO3** and biallelic activation of the maternally expressed non-coding RNAs **MEG3, MEG8, RTL1as** and the embedded miRNA/snoRNA cluster [PMID:

18471925;

P 25351781

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3. **Loss of DLK1** (a Notch-pathway non-canonical ligand) *leads to* dysregulated GnRH-axis timing and altered adipogenesis, *resulting in* **central precocious puberty** and **increased body fat** [PMID: 42677749]. (The precise circuit is still being resolved and appears to act up- or downstream of GnRH neurons — *inferred* from iPSC modelling [PMID: 40924042].)
4. Loss of paternal gene dosage across the domain (*DLK1/RTL1*) *results in* impaired fetal/placental growth signalling, *leading to* **pre- and postnatal growth restriction** and **small hands/feet** (*inferred* from mouse upd(12) and *Dlk1*-null models) [PMID: 17449025;

P 42677749

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5. Reduced muscle maturation and connective-tissue/skeletal effects *lead to* **neonatal hypotonia and feeding difficulties** (*inferred* from the mouse upd(12) muscle/bone phenotype) [PMID: 17449025].
6. Loss of DLK1's restraint on adipogenesis plus later positive energy balance *leads to* **truncal obesity**, which *results in* **dyslipidemia and type 2 diabetes** in later life (though hypercholesterolemia/diabetes can precede overt obesity) [PMID: 39693239;

P 26395259

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Where the mechanism **branches:** the identical structural deletion produces **TS14 if paternally inherited** but the mirror disorder **Kagami-Ogata syndrome if maternally inherited** — opposite dosage, opposite (overgrowth) phenotype [PMID: 39210340].

Supporting mechanistic detail

- **Molecular pathways.** DLK1 is a non-canonical Notch ligand; its activation during neuronal differentiation induces Wnt-pathway genes and alters neural fate, linking DLK1 dosage to neurodevelopment and pubertal timing [PMID: 40924042]. RTL1 is reciprocally regulated by RTL1as-encoded microRNAs: "these results suggest that RTL1as-encoded microRNA functions as a repressor of RTL1" [PMID: 22917972].
- **Cellular processes.** Skeletal-muscle maturation, bone formation, placental growth/organization, adipocyte differentiation, and GnRH-neuron ontogeny [PMID: 17449025;

P 40924042

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The domain's miRNAs (e.g., miR-127-3p) regulate cell-cycle arrest/senescence in model systems [PMID: 33279585].

- **Protein dysfunction.** Loss of function of DLK1 (and RTL1/DIO3 proteins). DIO3 is a deiodinase affecting local thyroid-hormone metabolism — a plausible contributor to growth/metabolic features (*inferred*).
- **Metabolic changes.** Increased adiposity, dyslipidemia, and early insulin resistance/T2DM [PMID: 39693239].
- **Suggested ontology terms.** GO: GO:0007219 (Notch signaling), GO:0045444 (fat cell differentiation), GO:0001701 (in utero embryonic development). CL: CL:0000136 (adipocyte), CL:0000187 (muscle cell), CL:0000617 (GnRH neuron, approximate). CHEBI: cholesterol (CHEBI:16113), glucose (CHEBI:17234).

Section 7 — Anatomical Structures Affected

- **Organ/system level.** Endocrine system (hypothalamic-pituitary-gonadal axis — precocious puberty; metabolic/endocrine — obesity, dyslipidemia, T2DM); musculoskeletal system (hypotonia, short stature, small hands/feet); nervous system (motor/language delay, mild ID); digestive (neonatal feeding difficulties); and the placenta prenatally. Fatty-liver disease has been reported as a newer-recognized feature [PMID: 33746039].
- **Tissue/cell level.** Skeletal muscle, bone/connective tissue, adipose tissue, and GnRH neurons are the principal targets (*inferred* from model organisms) [PMID: 17449025; P 40924042].
- **Subcellular level.** DIO3 and thyroid-hormone handling implicate cytoplasmic/membrane compartments; the central lesion is nuclear/chromatin (DMR methylation). GO cellular-component: GO:0005634 (nucleus, site of imprinting lesion).
- **Localization / lateralization.** Effects are systemic and bilateral/symmetric; body asymmetry (characteristic of SRS) is generally absent in TS14, aiding differentiation [PMID: 25351781].
- **Suggested UBERON terms.** UBERON:0001013 (adipose tissue), UBERON:0001134 (skeletal muscle), UBERON:0002107 (liver), UBERON:0001898 (hypothalamus), UBERON:0000992 (ovary), UBERON:0001987 (placenta).

Section 8 — Temporal Development

- **Onset.** Congenital/neonatal: growth restriction is prenatal; hypotonia and feeding difficulties present at birth. Onset is insidious/chronic rather than acute.
- **Progression.** Core dysmorphic/growth features are stable and non-progressive. Puberty becomes precocious in childhood (mean menarche ~10 years 2 months) [PMID: 24891339]. The metabolic phenotype is progressive and age-dependent — hypercholesterolemia from ≥6 years, diabetes from ≥9 years [PMID: 39693239].
- **Disease course.** Chronic and lifelong but compatible with normal life expectancy and favourable adult function [PMID: 39693239].
- **Critical periods / intervention windows.** Infancy/early childhood for feeding support and growth-hormone initiation (benefit shown with early start); pre-pubertal window for GnRH-analogue therapy; lifelong metabolic surveillance thereafter [PMID: 36977395;

P 39693239

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Section 9 — Inheritance and Population

- **Epidemiology.** Exact prevalence and incidence are **unknown**; TS14 is rare and recognized as under-diagnosed, detected in ~2% of patients referred for PWS or SRS testing [PMID: 29159982;

P 24891339

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- **Inheritance pattern.** Most cases are **sporadic/de novo** (upd(14)mat from maternal nondisjunction + trisomy rescue; sporadic epimutation). **Deletion-based cases are Mendelian with a parent-of-origin effect** — paternal transmission → TS14; maternal transmission of the same deletion → Kagami-Ogata syndrome [PMID: 39210340].
- **Penetrance / expressivity.** High penetrance for the core phenotype among those with the lesion; expressivity is variable. No genetic anticipation (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects / consanguinity.** No established founder effects; consanguinity is not a driver (this is not autosomal-recessive). Mosaic trisomy 14 can coexist with upd(14)mat [PMID: 30181735;

P 33746039

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- **Population demographics.** No strong ethnic/geographic predilection; cases are reported across European, Japanese, and Chinese populations [PMID: 39693239; P 42026594].
  No consistent sex bias is established, although precocious-puberty ascertainment skews some cohorts toward girls.

Section 10 — Diagnostics

Tiered molecular approach (all three mechanisms share MEG3-DMR hypomethylation):

1. **First-line — methylation analysis** of the MEG3/DLK1 IG-DMR and MEG3-DMR by methylation-specific MLPA (MS-MLPA; ME032/ME034 kits) or bisulfite pyrosequencing detects the hypomethylation common to all three mechanisms [PMID: 40229547; P 25351781].
2. **Microsatellite / SNP-array genotyping** of parent–child trios distinguishes upd(14)mat [PMID: 33746039].
3. **MLPA / chromosomal microarray / FISH** detect paternal 14q32 microdeletions [PMID: 40229547].

Prenatal diagnosis. NIPT followed by amniotic MS-MLPA/CMA/FISH can detect imprinted 14q32 abnormalities, including recurrent KOS/TS14 in at-risk families [PMID: 40229547; P 39210340].

Clinical/laboratory findings. Lab monitoring for lipids and glucose given cardiometabolic risk; endocrine evaluation (LH/FSH, bone age) for precocious puberty.

Differential diagnosis. Silver-Russell syndrome (11p15/upd(7)mat), Prader-Willi syndrome (15q11-q13), and Schaaf-Yang syndrome (*MAGEL2*), distinguished by locus-specific methylation testing. "Most patients with 14q32.2 disruption fulfill the criteria for a clinical diagnosis of SRS" [PMID: 29659920], and "The most common genetic diagnoses were Temple syndrome (formerly known as maternal uniparental disomy 14), Schaaf-Yang syndrome..." in PWS-like cohorts [PMID: 34460908]. Testing for TS14 is recommended when PWS/SRS testing is negative [PMID: 36362517].

Section 11 — Outcome / Prognosis

Survival and cognition are generally favourable. Life expectancy is essentially normal; there are no TS14-specific mortality data indicating reduced survival. In the 60-patient Japanese cohort, "42.9% of patients were enrolled in special classes in childhood, whereas 98.2% of patients attended college or had jobs in adulthood" [PMID: 39693239]. Median IQ is ~85–91 [PMID: 36362517].

Morbidity is cardiometabolic. Long-term risks: intellectual/developmental disability 21.6%, neurodevelopmental disorders 21.6%, obesity 20.0%, hypercholesterolemia 26.5% (≥6 y), and diabetes mellitus 12.8% (≥9 y) [PMID: 39693239]. Importantly, hypercholesterolemia and diabetes sometimes preceded obesity and were controllable with oral medication, underscoring the need for early metabolic surveillance [PMID: 39693239; P 26395259].

Prognostic factors. Earlier diagnosis permits earlier GH/GnRH-analogue intervention and metabolic monitoring, improving growth and (likely) cardiometabolic outcomes. Final adult height without treatment is reduced (median ~ −2.04 SDS) [PMID: 24891339].

Section 12 — Treatment

Management is **supportive, multidisciplinary, and non-curative** — no disease-modifying or gene-directed therapy exists.

Intervention	Evidence / effect	Source	NCIT (suggested)
Recombinant growth hormone (rhGH)	1 yr: height +1.31 SDS, HV 5.30 cm/yr. 5 yr: height SDS −1.78 → +0.11; fat-mass% SDS fell, lean-body-mass SDS rose	P 30836360; P 36977395	Recombinant Human Growth Hormone (NCIT:C1762)
GnRH-analogue therapy	Ameliorated central precocious puberty in 32/60 patients	P 39693239	GnRH Analogue
Neonatal tube/ supportive feeding	For feeding difficulties/failure to thrive	P 26395259	Enteral Nutrition (NCIT:C15190)
Physiotherapy / OT / speech therapy	For hypotonia, motor and language delay	P 36362517	Rehabilitation Therapy (NCIT:C15917)
Metabolic management (statins/ oral hypoglycemics, diet)	Dyslipidemia/early T2DM controllable with oral medication	P 39693239	—

GH response in TS14 is "similar to the response in patients treated under the small for gestational age indication" [PMID: 30836360]. Over 5 years, "mean (95% CI) height SDS increased significantly during 5 years of GH treatment from -1.78 (-2.52; -1.04) to 0.11 (-0.66; 0.87). Fat mass percentage SDS decreased significantly during the first year of GH, and lean body mass (LBM) SDS and LBM index increased significantly" [PMID: 36977395]. No pharmacogenomic, gene-therapy, cell-therapy, RNA-based, immunotherapy, or TS14-specific surgical treatments are established.

Section 13 — Prevention

- **Primary prevention.** None available for the genetic lesion. Counselling about advanced-maternal-age nondisjunction risk and ART-associated imprinting-defect risk is relevant [PMID: 30181735; P 41153459].

- **Secondary prevention.** Early recognition via testing TS14 in PWS/SRS-like infants with negative primary testing [PMID: 36362517]; prenatal detection in deletion-carrier families (NIPT + confirmatory MS-MLPA/CMA/FISH) [PMID: 40229547].
- **Tertiary prevention.** Lifelong surveillance and management of growth and cardiovascular risk factors, as explicitly recommended: "improved diagnosis and long-term monitoring, especially of growth and cardiovascular risk factors, is required" [PMID: 24891339].
- **Genetic counselling.** Essential for deletion carriers given the parent-of-origin recurrence of TS14 vs KOS [PMID: 39210340]. No vaccine or public-health/environmental prevention applies.

Section 14 — Other Species / Natural Disease

- **Taxonomy / orthology.** The human 14q32 domain is syntenic with **mouse distal chromosome 12** (*Mus musculus*, NCBI Taxon 10090). Orthologues: *Dlk1*, *Rtl1/Peg11*, *Dio3*, *Meg3/Gtl2*, *Meg8*, *Rian/Mirg* [PMID: 18471925].
- **Natural disease in animals.** No well-characterized naturally occurring Temple-syndrome analogue is documented in companion animals or wildlife (no evidence found).
- **Comparative biology / evolutionary conservation.** The imprinted architecture and parent-of-origin regulation of the DLK1-DIO3 domain are conserved between human and mouse, making the mouse a valid model of the dosage imbalance [PMID: 18471925; P 17449025].
- **Transmission.** Not applicable — TS14 is non-infectious and non-zoonotic.

Section 15 — Model Organisms

- **Mouse uniparental disomy chromosome 12 [upd(12)].** Reproduces the TS14-type dosage imbalance and causes "defects in skeletal muscle maturation, bone formation, placenta size and organization and prenatal lethality" [PMID: 17449025] — recapitulating the muscle/growth/placental aspects of TS14; prenatal lethality is a limitation not seen in human TS14.
- **Dlk1-knockout mouse.** Shows growth retardation, increased adiposity/obesity, and metabolic changes recapitulating core TS14 features; a recent study found "Dlk1-deficient

mice achieve puberty despite low body weight and low levels of leptin and kisspeptin," informing the DLK1–puberty link [PMID: 42677749].

- **Human iPSC / stem-cell models.** hPSC *DLK1*-edited lines reveal a dual role of DLK1 in GnRH neuron ontogeny, with timing-dependent effects on GnRH lineage commitment [PMID: 40924042]. iPSC models of imprinting disorders are feasible but prone to culture-induced hypermethylation of the MEG3/DLK1:IG-DMR unless specialized media/hypoxic conditions are used [PMID: 36578048].
- **In-vitro domain biology.** Mouse chromosome-12 miRNA (e.g., miR-127-3p) studies illuminate cell-cycle/senescence regulation by the cluster [PMID: 33279585].
- **Limitations.** Full upd(12) mouse shows prenatal lethality; single-gene (*Dlk1*) models capture the metabolic/pubertal axis but not the full multi-gene imbalance; iPSC models require careful methylation control.
- **Resources.** MGI (mouse), Alliance of Genome Resources.

Mechanistic Model / Interpretation

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Paternal 14q32 lesion (upd(14)mat / IG-DMR epimutation / paternal deletion)
                              |
                              v
       Loss of paternal IG-DMR methylation  --> unified biomarker: MEG3-DMR hypomethylation
                              |
       +----------------------+----------------------+
       v                                             v
 Silence paternal genes                    Activate maternal ncRNAs (biallelic)
  DLK1 v , RTL1 v , DIO3 v                  MEG3 ^ , MEG8 ^ , RTL1as ^ , miRNAs ^
       |                                             |
       v                                             v
 DLK1 loss -> GnRH-axis dysregulation       RTL1as miRNA represses RTL1 (reinforces loss)
 + increased adipogenesis                   -> placental/growth effects
       |
       +--> Central precocious puberty (86-89%)
       +--> Pre/postnatal growth restriction, small hands/feet
       +--> Hypotonia, feeding difficulties (muscle immaturity)
       +--> Truncal obesity -> dyslipidemia -> early T2DM

PARENT-OF-ORIGIN BRANCH (deletion carriers):
  Paternal transmission --> TS14 (loss of paternal program)
  Maternal transmission --> Kagami-Ogata syndrome (mirror / overgrowth)
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The unifying insight is **dosage reciprocity**: the three mechanisms converge on the same methylation biomarker and the same net dosage imbalance, while the DLK1-DIO3 domain's sensitivity to parent-of-origin makes TS14 and KOS mechanistic mirror images.

Evidence Base (key literature)

PMID	Role
24891339	Eponym origin; pooled phenotype frequencies & growth metrics from 51 cases
32592473	Defines 14q32 locus, parent-of-origin mechanisms, TS14↔KOS relationship
38715103	Three mechanisms; gene-dosage imbalance; clinical features
29659920	Mechanism frequencies; SRS overlap
39693239	Largest cohort (n=60) — phenotype frequencies, morbidity, outcome, treatment
30836360	rhGH 1-year growth response
36977395	Long-term (5-yr) GH/body-composition benefit
42026594	Chinese cohort phenotype & GH therapy
42677749	DLK1 LOF → CPP & adiposity (mouse)
40924042	DLK1 mechanism in GnRH neuron ontogeny; iPSC model
18471925	Domain architecture; IG-DMR hierarchy
17449025	Mouse upd(12) model phenotype
22917972	RTL1as miRNA represses RTL1
39210340	Parent-of-origin recurrence (TS14 mother → KOS offspring)
31014393	Novel template-switching mechanism for UPD(14)mat
29159982	Under-diagnosis rate (2.1% of PWS referrals)
40229547	Diagnostic modalities (MS-MLPA, FISH, CMA, NIPT)
34460908	Synonym (maternal UPD14); PWS-like differential dx
36362517	Body composition, IQ, testing recommendation (n=15)
26395259	Isolated epimutation case; metabolic risk
41153459	ART epimutation risk
35266280	Multilocus imprinting disturbance (11p15 + 14q32)
30234781	Gene–environment interaction (preeclampsia, DLK1-MEG3)
30181735	Trisomy-rescue origin (TS14 + mosaic trisomy 14)

Limitations and Knowledge Gaps

1. **Epidemiology is undefined.** No reliable prevalence or incidence figures exist; TS14 is recognized as under-diagnosed [PMID: 24891339; P 29159982].
2. **Ascertainment bias.** Reported mechanism proportions and phenotype frequencies vary widely between cohorts (e.g., upd(14)mat ~29% vs ~52%) because referral routes (precocious puberty vs growth clinics vs PWS/SRS testing) differ [PMID: 29659920; P 39693239].
3. **Mechanistic attribution is partly inferred.** The specific contribution of each gene (DLK1 vs RTL1 vs DIO3) to each clinical feature relies heavily on mouse/iPSC models; the exact pathway from DLK1 loss to CPP remains incompletely resolved and appears to act up-/downstream of GnRH neurons [PMID: 40924042].
4. **Treatment evidence is limited and short-to-medium term.** GH and GnRH-analogue data derive from small, retrospective, uncontrolled cohorts; long-term adult cardiometabolic and final-height outcomes with treatment are not fully quantified.
5. **No disease-modifying therapy or validated biomarkers of progression** exist, and there are no TS14-specific quality-of-life instruments.
6. **Sex-ratio and population-genetic data are sparse**, and there are no natural animal-disease analogues.

Proposed Follow-up Experiments / Actions

1. **Establish a prospective international TS14 registry** with uniform molecular and phenotyping protocols to resolve true prevalence, mechanism-specific genotype–phenotype correlations, and long-term cardiometabolic outcomes.
2. **Controlled/longitudinal treatment studies** of rhGH and GnRH-analogue therapy with adult final-height, body-composition, lipid, and glucose endpoints; test whether early GH initiation (<2 y) modifies metabolic trajectory.

3. **Dissect DLK1's CPP mechanism** using conditional/temporally controlled DLK1 models and patient iPSC-derived hypothalamic/GnRH-neuron organoids to define whether the pubertal defect is pre- or post-synaptic to GnRH neurons [building on

   P 40924042

   ].
4. **Define metabolic-surveillance thresholds**: prospectively characterize the onset sequence of hypercholesterolemia, obesity, and T2DM to set evidence-based screening intervals, given that lipids/glucose can precede obesity [PMID: 39693239].
5. **Quantify ART-associated epimutation risk specifically for 14q32** in large ART-conceived cohorts to inform preconception and prenatal counselling [PMID: 41153459].
6. **Develop methylation-based multi-locus diagnostic panels** (and refine NIPT workflows) to improve early/prenatal detection and reduce diagnostic delay, especially in PWS/SRS-like infants with negative first-line testing [PMID: 40229547;

   P 36362517

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Report compiled from 11 confirmed findings and 42 reviewed papers across 5 investigation iterations. Evidence types: human clinical cohorts/case series (primary), model-organism (mouse upd(12), Dlk1-KO), and in-vitro (iPSC, placental) studies.
